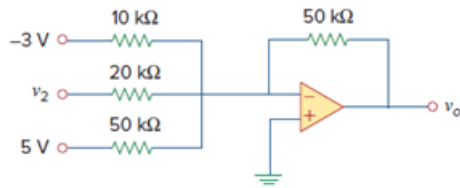


Problem

For the op amp circuit in Fig. 5.76, determine the value of v_2 in order to make $v_o = -16.5$ V.

Figure 5.76:



Step-by-step solution

Step 1 of 2

Refer to Figure 5.76 in the text book for the op-amp summing amplifier circuit.

The output of the summing amplifier is,

$$v_o = \left(-\frac{R_f}{R_1} v_1 - \frac{R_f}{R_2} v_2 - \frac{R_f}{R_3} v_3 \right)$$

$$v_o = -R_f \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} \right)$$

Step 2 of 2

Determine the voltage, v_2 by substituting 3 V for v_1 , 16.5 V for v_o , 5 V for v_3 , 50 k Ω for R_f , 10 k Ω for R_1 , 20 k Ω for R_2 and 50 k Ω for R_3 .

$$-16.5 = -(50 \text{ k}\Omega) \left(-\frac{3}{10 \text{ k}\Omega} + \frac{v_2}{20 \text{ k}\Omega} + \frac{5}{50 \text{ k}\Omega} \right)$$

$$0.33 = 0.05v_2 - 0.2$$

$$v_2 = 10.6 \text{ V}$$

Thus, the value of the voltage v_2 is 10.6 V.